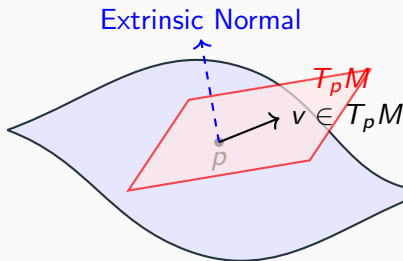


Differential Geometry and Topology in Time Series

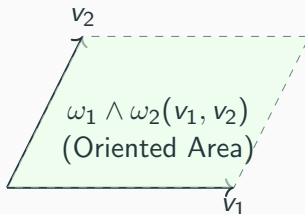
Smooth Manifolds and Phase Space

- **Smooth Manifold (M):** Locally homeomorphic to Euclidean space. Permits global differential calculus.
- **Tangent Space ($T_p M$):** Vector space of all directional velocities at $p \in M$.
- **Takens' Embedding:** Univariate time series embedded into a higher-dimensional phase space. Deterministic dynamics constrain vectors to a low-dimensional topological manifold.



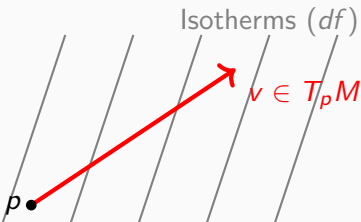
Tensors and the Wedge Product

- **$(0, k)$ -Tensor:** Covariant multilinear map
 $\omega : T_p M \times \cdots \times T_p M \rightarrow \mathbb{R}$.
- **Total Antisymmetry:**
 $\omega(\dots, v_i, \dots, v_j, \dots) = -\omega(\dots, v_j, \dots, v_i, \dots)$. Output is strictly zero for linearly dependent vectors.
- **Wedge Product $(\wedge)$:** $\alpha \wedge \beta = (-1)^{kl} \beta \wedge \alpha$. Constructs higher-degree forms computing oriented volumes. $\omega \wedge \omega = 0$.



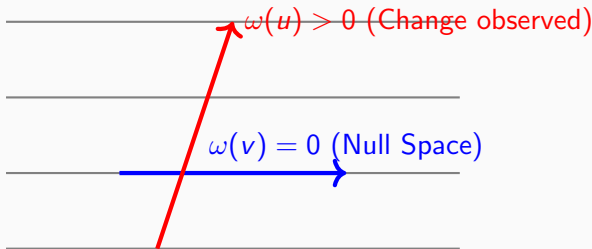
Tangent and Cotangent Bundles

- **Tangent Bundle (TM):** Disjoint union of all tangent spaces. Vectors represent physical velocities and mathematically valid state transitions.
- **Cotangent Bundle (T^*M):** Disjoint union of all formal dual vector spaces T_p^*M . Elements are 1-forms $\omega : T_pM \rightarrow \mathbb{R}$.
- **Time Series Application:** 1-forms act as sensors/observables. The dual pairing $df(v)$ computes the instantaneous rate of change of the observable f along trajectory v .



Mechanics of 1-Forms

- **Geometric Structure:** A 1-form is a family of parallel, equally spaced hyperplanes. Evaluation $\omega(v)$ counts pierced hyperplanes.
- **Null Space (Kernel):** Subspace where $\omega(v) = 0$. Represents state transitions where the observable remains strictly constant.



Riemannian Geometry

- **Metric Tensor (g):** Symmetric, positive-definite bilinear form defining local inner products. Enables dynamic scaling of distances across phase space.
- **Levi-Civita Connection:** Unique affine connection preserving the metric ($\nabla g = 0$) and strictly torsion-free ($\nabla_X Y - \nabla_Y X = [X, Y]$).
- **Covariant Derivative (∇):** Intrinsic rate of change. Projects variation onto $T_p M$, discarding apparent coordinate curvature.

Parallel Transport ($\nabla_X v = 0$)



The Laplace-Beltrami Operator

- **Definition:** Generalisation of the Euclidean Laplacian to Riemannian manifolds.

$$\Delta_g f = \operatorname{div}(\operatorname{grad} f) = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} g^{ij} \partial_j f \right)$$

- **Dynamics:** Infinitesimal generator of Brownian motion on the manifold.
- **Diffusion Maps:** Eigenfunctions of Δ_g provide optimal, non-linear coordinates invariant to isometric deformations.

- **Exterior Derivative (d):** $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$. Satisfies $d^2 = 0$. Computes structural gradients across phase space.
- **Hodge Star ($\star$):** Isomorphism $\star : \Omega^k(M) \rightarrow \Omega^{n-k}(M)$. Maps tangent dynamics to orthogonal complements using the metric g .
- **Codifferential (δ):** Adjoint of d . $\delta = (-1)^{nk+n+1} \star d \star$. Computes geometric divergence/convergence of feature fields.
- **Hodge Laplacian:** $\Delta_k = d_{k-1} \delta_k + \delta_{k+1} d_k$.

Topology and De Rham Cohomology

- **Closed Forms:** $d\omega = 0$.
- **Exact Forms:** $\omega = d\eta$. Since $d^2 = 0$, exact $\implies$ closed.
- **De Rham Cohomology ($H_{dR}^k(M)$):** Quotient space of closed forms modulo exact forms. Betti numbers compute macroscopic holes.
- **Time Series:** $H_{dR}^1(M)$ detects fundamental periodic cycles invariant to noise and deformations.

